Supplemental Table 1. PRISMA checklist.

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	1-2
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	2
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	2
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	2
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	2 and Suppl. Table 2

Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	2 and 5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5 and Table 1
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5-6 and Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	5-6 and Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	5 and Table 1
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	-
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of	7-10 and Figures

		consistency.	2-5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10, 17, Table 3 and Figures 6-8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10 and Table 2
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	17-18, 20 and 23-24
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	24
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	24 and Figure 9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	25

Supplemental Table 2. Search strategy for each database (until May 31st, 2020 included).

DATABASE	**SEARCH STRATEGY**
Cochrane Library	((“HCC” OR “hepatocarcinoma” OR “hepatocellular carcinoma”) AND (“HIF” OR “hypoxia-inducible factor”)):ti,ab,kw
Embase	(‘hcc’ OR ‘hepatocarcinoma’ OR ‘hepatocellular carcinoma’) AND (‘hif’ OR ‘hypoxia-inducible factor’)
PubMed	(“HCC”[All Fields] OR “hepatocarcinoma”[All Fields] OR “hepatocellular carcinoma”[All Fields]) AND (“HIF”[All Fields] OR “hypoxia-inducible factor”[All Fields])
Scopus	TITLE-ABS-KEY (("HCC" OR "hepatocarcinoma" OR "hepatocellular carcinoma") AND ("HIF" OR "hypoxia-inducible factor"))
Web of Science Core Collection	TS=(("HCC" OR "hepatocarcinoma" OR "hepatocellular carcinoma") AND ("HIF" OR "hypoxia-inducible factor")) *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC* *Timespan=All years*

Supplemental Table 3. Antibodies and staining procedure employed for immunohistochemistry.

Study (reference)	Year	Antibody	Clone and Source	Dilution	Incubation conditions
Huang et al.[30]	2005	Monoclonal anti-HIF-1α	Clone H1alpha67, Neomarkers	NR	4ºC overnight
Wada et al.[31]	2006	Mouse monoclonal anti-HIF-1α	Mouse monoclonal IgG2a, Novus Biologicals, Inc., Littleton, CO	1:50	4ºC overnight
Xie et al.[32]	2008	Monoclonal anti-HIF-1α	Cayman Chemical, Ann Arbor, MI, USA	1:100	4ºC overnight
Dai et al.[33]	2009	Rabbit polyclonal anti-HIF-1α	ab65979, Abcam, Cambridge, UK	1:50	NR
Liu et al.[34]	2010	Anti-HIF-1α	Sigma-Aldrich, MI, USA	NR	NR
Xiang et al.[35]	2011	Mouse monoclonal anti-human anti-HIF-1α	Santa Cruz Biotechnology, CA, USA	NR	4ºC overnight
Li et al.[36]	2012	Mouse anti-human monoclonal anti-HIF-1α	England NeoMarker Company, England	NR	NR
Xia et al.[37]	2012	Anti-HIF-1α	Sigma-Aldrich, MI, USA	NR	4ºC overnight
Xiang et al.[38]	2012	Mouse monoclonal anti-human anti-HIF-1α	Santa Cruz Biotechnology, CA, USA	NR	4ºC overnight
Cui et al.[39]	2013	Anti-HIF-1α	Santa Cruz Biotechnology, CA, USA	NR	37ºC for 1 h
Ma et al.[40]	2013	Rabbit monoclonal anti-human anti-HIF-1α	ab51608, Abcam, American Abcam Company, USA	1:100	NR
Wang et al.[41]	2014	Mouse monoclonal anti-human anti-HIF-1α	sc-13515, Santa Cruz Biotechnology, CA, USA	NR	1-2 h
Yang et al.[42]	2014	Monoclonal anti-HIF-1α	Clone H1alpha67, Santa Cruz Biotechnology, CA, USA	NR	37ºC for 1 h + 4ºC overnight
Huang et al.[43]	2015	Rabbit monoclonal anti-HIF-1α	Abcam, Cambridge, UK	NR	37ºC for 1 h
Li et al.[44]	2015	Monoclonal anti-HIF-1α	sc-13515, Santa Cruz Biotechnology, CA, USA	1:200	4ºC overnight
Srivastava et al.[45]	2015	Mouse monoclonal anti-HIF-1α	Clone H1alpha67, Novus Biological, USA	1:1000	NR
Zhao et al.[46]	2015	Anti-HIF-1α	NR	NR	4ºC overnight
Tang et al.[47]	2016	Rabbit polyclonal anti-HIF-1α	Abcam, Cambridge, UK	NR	NR
Wang et al.[48]	2017	Rabbit monoclonal anti-HIF-1α	ZA-0562, USA	NR	4ºC overnight

Dai et al.[49]	2018	Mouse monoclonal anti-HIF-1α	Clone H1alpha67, Santa Cruz Biotechnology, CA, USA	NR	NR
Tian et al.[50]	2018	Rabbit anti-human anti-HIF-1α	Wanjiang Biotechnology Co., Ltd., Shanghai, China	NR	NR
Wang et al.[51]	2018	Anti-HIF-1α	Abcam, Cambridge, MA, USA	NR	4ºC overnight
Zou et al.[52]	2018	Rabbit monoclonal anti-HIF-1α	Boster, PB0245, China	1:200	4ºC for 12 h
Gong et al.[53]	2019	Anti-HIF-1α	A11945, ABclonal, USA	NR	4ºC overnight
Wu et al.[54]	2019	Anti-HIF-1α	ab2185, Abcam, Cambridge, MA, USA	NR	4ºC overnight
Qian et al.[56]	2020	Anti-HIF-1α	GB111339, Wuhan Servicebio, China	1:200	4º overnight
Study (Reference)	**Year**	**Antibody**	**Clone and Source**	**Dilution**	**Incubation conditions**
Bangoura et al.[57]	2004	Mouse monoclonal anti-HIF-2α	Anti-HIF-2α/EPAS1 (190b), DAKO, Glostrup, Denmark	NR	NR
Bangoura et al.[58]	2007	Mouse monoclonal anti-HIF-2α	Anti-HIF-2α/EPAS1 (190b), Santa Cruz Biotechnology, CA, USA	1:80	RT for 1 h
Sun et al.[59]	2013	Anti-HIF-2α	ab199, Abcam, Cambridge, MA, USA	1:200	RT for 1 h
Yang et al.[42]	2014	Polyclonal anti-HIF-2α	Boster Biological Technology, Wuhan, China	NR	37ºC for 1 h + 4ºC overnight
Yang et al.[60]	2016	Polyclonal anti-HIF-2α	ChIP Grade ab199, Abcam, Cambridge, MA, USA	NR	NR
		Mouse monoclonal anti-HIF-2α	Anti-HIF-2α/EPAS1 (190b), Santa Cruz Biotechnology, CA, USA	NR	NR
Jiang et al.[61]	2018	Anti-HIF-2α	ab8365, Abcam, Cambridge, UK	1:150	4ºC overnight
Chen et al.[62]	2019	Anti-HIF-2α	7096, Cell Signaling Technology	1:300	4ºC for 8 h

NR, not reported; RT, room temperature